

General Relativity HW-2

Remark: the connections that appear in this homework are all affine connections.

PROBLEM 1. Let us have a sphere S^2 in Cartesian coordinates $\{x^1, x^2, x^3\}$ such that $(x^1)^2 + (x^2)^2 + (x^3)^2 = 1$. Then we do stereographically project it on a plane $x^3 = -1$ which has coordinates $\{y^1, y^2\}$ and on a plane $x^3 = 1$ which has coordinates $\{z^1, z^2\}$. Explicit maps are:

$$\phi_y(x^i) \equiv (y^1, y^2) = \left(\frac{2x^1}{1-x^3}, \frac{x^2}{1-x^3} \right)$$

and

$$\phi_z(x^i) \equiv (z^1, z^2) = \left(\frac{2x^1}{1+x^3}, \frac{2x^2}{1+x^3} \right)$$

Find $\phi_z \circ \phi_y^{-1}$, i.e. $z^k(y^j)$.

SOLUTION:

Firstly, in order to calculate ϕ_y^{-1} , one has:

$$\begin{cases} y^1 = \frac{2x^1}{1-x^3} \\ y^2 = \frac{2x^2}{1-x^3} \\ (x^1)^2 + (x^2)^2 + (x^3)^2 = 1 \end{cases} \implies \frac{(1-x^3)^2}{4}((y^1)^2 + (y^2)^2) + (x^3)^2 = 1$$

Through some simplification we have:

$$\frac{1-x^3}{4}((y^1)^2 + (y^2)^2) = 1+x^3 \implies x^3 = \frac{((y^1)^2 + (y^2)^2) - 4}{((y^1)^2 + (y^2)^2) + 4}$$

Then let's transform it into z :

$$z^1 = \frac{2x^1}{1+x^3} = y^1 \frac{1-x^3}{1+x^3} = y^1 \frac{4}{((y^1)^2 + (y^2)^2) + 4}$$
$$z^2 = \frac{2x^2}{1+x^3} = y^2 \frac{1-x^3}{1+x^3} = y^2 \frac{4}{((y^1)^2 + (y^2)^2) + 4}$$

which solves the problem.

PROBLEM 2. Given there are two connections Γ and $\hat{\Gamma}$. Show that

$$S_{\mu\nu}^{\rho} = \Gamma_{\mu\nu}^{\rho} - \hat{\Gamma}_{\mu\nu}^{\rho}$$

is a tensor.

SOLUTION:

Solution 1: If we define connection in the way that

$$\mathcal{D}_{\mu} V^{\rho} = \partial_{\mu} V^{\rho} + \Gamma_{\mu\nu}^{\rho} V^{\nu}$$

is a tensor ($\forall$ tensor V^{ρ}), then it is apparent that $S_{\mu\nu}^{\rho} = \Gamma_{\mu\nu}^{\rho} - \hat{\Gamma}_{\mu\nu}^{\rho}$ is a tensor because tensors (with the same covariant, contravariant order) are closed under addition, thus

$$S_{\mu\nu}^{\rho} V^{\nu} = \mathcal{D}_{\mu} V^{\rho} - \hat{\mathcal{D}}_{\mu} V^{\rho}$$

is a tensor, and then $S_{\mu\nu}^{\rho}$ is a tensor because V^{ν} is also a tensor. $\square$

Solution 2: If we define connections abstractly, by which I mean the operators on the possible sections of tangent bundle of a manifold which obeys Leibniz rules, then suppose we have the following local coordinate representation:

$$\nabla_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}} = \Gamma_{\mu\nu}^{\rho} \frac{\partial}{\partial x^{\rho}}, \hat{\nabla}_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}} = \hat{\Gamma}_{\mu\nu}^{\rho} \frac{\partial}{\partial x^{\rho}}$$

where the index μ represents the direction we are taking directional derivative, ν represents the tangent vector we are taking derivative. Then we have Leibniz rules:

$$\begin{aligned} \nabla_{f \frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}} &= f \nabla_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}} \\ \nabla_{\frac{\partial}{\partial x^{\mu}}} (f \frac{\partial}{\partial x^{\nu}}) &= \frac{\partial f}{\partial x^{\mu}} \frac{\partial}{\partial x^{\nu}} + f \nabla_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}} \end{aligned}$$

for any scalar field f on the manifold. When we consider a coordinate transformation $x \rightarrow q$, then:

$$\nabla_{\frac{\partial}{\partial q^{\mu'}}} (\frac{\partial}{\partial q^{\nu'}}) = \nabla_{\frac{\partial x^{\mu}}{\partial q^{\mu'}} \frac{\partial}{\partial x^{\mu}}} (\frac{\partial x^{\nu}}{\partial q^{\nu'}} \frac{\partial}{\partial x^{\nu}}) = \frac{\partial x^{\mu}}{\partial q^{\mu'}} [(\frac{\partial}{\partial x^{\mu}} (\frac{\partial x^{\nu}}{\partial q^{\nu'}})) \frac{\partial}{\partial x^{\nu}} + \frac{\partial x^{\nu}}{\partial q^{\nu'}} \nabla_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}}]$$

from here we see that ∇ is not a tensor. But when considering $\nabla - \hat{\nabla}$, we find that the first non-covariant term just disappear and then:

$$\nabla_{\frac{\partial}{\partial q^{\mu'}}} (\frac{\partial}{\partial q^{\nu'}}) - \hat{\nabla}_{\frac{\partial}{\partial q^{\mu'}}} (\frac{\partial}{\partial q^{\nu'}}) = (\frac{\partial x^{\mu}}{\partial q^{\mu'}}) (\frac{\partial x^{\nu}}{\partial q^{\nu'}}) (\nabla_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}} - \hat{\nabla}_{\frac{\partial}{\partial x^{\mu}}} \frac{\partial}{\partial x^{\nu}})$$

which precisely implies that $\nabla - \hat{\nabla}$ is a tensor. $\square$

PROBLEM 3. Given a connection Γ show that torsion

$$T_{\mu\nu}^{\rho} = \Gamma_{\mu\nu}^{\rho} - \Gamma_{\nu\mu}^{\rho}$$

is indeed a tensor.

SOLUTION:

Consider a coordinate transformation $x \rightarrow q$. If we define connection in the way that

$$\mathcal{D}_{\mu} V^{\rho} = \partial_{\mu} V^{\rho} + \Gamma_{\mu\nu}^{\rho} V^{\nu}$$

is a tensor ($\forall$ tensor V^{ρ}), then

$$(\partial_{\mu'} V^{\rho'} + \Gamma_{\mu'\nu'}^{\rho'} V^{\nu'}) V^{\mu'} = \frac{\partial q^{\rho'}}{\partial x^{\rho}} [(\partial_{\mu} V^{\rho} + \Gamma_{\mu\nu}^{\rho} V^{\nu}) V^{\mu}]$$

swapping indices μ, ν we have

$$(\partial_{\nu'} V^{\rho'} + \Gamma_{\nu'\mu'}^{\rho'} V^{\mu'}) V^{\nu'} = \frac{\partial q^{\rho'}}{\partial x^{\rho}} [(\partial_{\nu} V^{\rho} + \Gamma_{\nu\mu}^{\rho} V^{\mu}) V^{\nu}]$$

subtracting two equations gives:

$$(V^{\mu'} \partial_{\mu'} V^{\rho'} - V^{\nu'} \partial_{\nu'} V^{\rho'}) + (\Gamma_{\mu'\nu'}^{\rho'} - \Gamma_{\nu'\mu'}^{\rho'}) V^{\mu'} V^{\nu'} = \frac{\partial q^{\rho'}}{\partial x^{\rho}} [(V^{\mu} \partial_{\mu} V^{\rho} - V^{\nu} \partial_{\nu} V^{\rho}) + (\Gamma_{\mu\nu}^{\rho} - \Gamma_{\nu\mu}^{\rho}) V^{\mu} V^{\nu}]$$

Thus to prove $T_{\mu\nu}^{\rho}$ is a tensor, it suffices to prove that $V^{\mu} \partial_{\mu} V^{\rho} - V^{\nu} \partial_{\nu} V^{\rho}$ is a tensor.

By using the same coordinate transformation above, we have:

$$\begin{aligned} V^{\mu'} \partial_{\mu'} V^{\rho'} - V^{\nu'} \partial_{\nu'} V^{\rho'} &= V^{\mu} \partial_{\mu} \left(\frac{\partial q^{\rho'}}{\partial x^{\rho}} V^{\rho} \right) - V^{\nu} \partial_{\nu} \left(\frac{\partial q^{\rho'}}{\partial x^{\rho}} V^{\rho} \right) \\ &= \frac{\partial q^{\rho'}}{\partial x^{\rho}} (V^{\mu} \partial_{\mu} V^{\rho} - V^{\nu} \partial_{\nu} V^{\rho}) + \left[\partial_{\mu} \left(\frac{\partial q^{\rho'}}{\partial x^{\rho}} \right) V^{\mu} V^{\rho} - \partial_{\nu} \left(\frac{\partial q^{\rho'}}{\partial x^{\rho}} \right) V^{\nu} V^{\rho} \right] \end{aligned}$$

where the latter term in middle bracket is simply 0 by substituting μ with ν because they are dummy indices. It is obvious that $V^{\mu} \partial_{\mu} V^{\rho} - V^{\nu} \partial_{\nu} V^{\rho}$ is a tensor, and we complete the proof. $\square$

PROBLEM 4. Show that for a Christoffel connection $\Gamma_{\mu\nu}^{\rho}$ one has

$$\Gamma_{\mu\nu}^{\mu} = \frac{1}{\sqrt{|g|}} \partial_{\nu} \sqrt{|g|}$$

where g is the determinant of a metric $g_{\mu\nu}$.

SOLUTION:

We begin the proof by considering the exact representation of $\partial_\nu \sqrt{|g|}$.

Due to the chain rule of taking derivatives, we would notice that:

$$\partial_\nu \sqrt{|g|} = \frac{1}{2\sqrt{|g|}} \frac{\partial |g|}{\partial g_{\mu\rho}} \partial_\nu g_{\mu\rho}$$

Notice that by the definition of determinant (and the fact that it is an anti-symmetric linear form), $\frac{\partial |g|}{\partial g_{\mu\rho}}$ is actually $M^{\mu\rho}$, which means the cofactor of $g_{\mu\rho}$ in linear algebra (the determinant of the matrix g after you remove the μ th row and the ρ th column, and then times $(-1)^{\mu+\rho}$)

By the definition of Levi-Civita connection, we have:

$$\partial_\nu g_{\mu\rho} = \Gamma_{\mu\nu}^\lambda g_{\lambda\rho} + \Gamma_{\rho\nu}^\lambda g_{\mu\lambda}$$

plug this into the above equation gives you that:

$$\partial_\nu \sqrt{|g|} = \frac{1}{2\sqrt{|g|}} M^{\mu\rho} (\Gamma_{\mu\nu}^\lambda g_{\lambda\rho} + \Gamma_{\rho\nu}^\lambda g_{\mu\lambda})$$

Now here is the tricky part of the proof. We change the sequence of summation so that we let $M^{\mu\rho} g_{\lambda\rho}$ (and $M^{\mu\rho} g_{\mu\lambda}$) to be firstly summed. But we notice that $\forall \lambda \neq \mu, M^{\mu\rho} g_{\lambda\rho}$ is actually the determinant of g' , where you substitute the μ th row of g with the λ th row of g . But we all know that if a matrix has two rows or two columns that are the same, then the determinant is 0! When $\mu = \lambda$, then $M^{\mu\rho} g_{\lambda\rho}$ is just the determinant of g by clear definition. Thus, $M^{\mu\rho} g_{\lambda\rho} = \delta_\lambda^\mu |g|$ and $M^{\mu\rho} g_{\mu\lambda} = \delta_\lambda^\rho |g|$.

Plugging the results into the above equation gives:

$$\partial_\nu \sqrt{|g|} = \frac{1}{2\sqrt{|g|}} (\Gamma_{\mu\nu}^\lambda \delta_\lambda^\mu |g| + \Gamma_{\rho\nu}^\lambda \delta_\lambda^\rho |g|) = \sqrt{|g|} \Gamma_{\mu\nu}^\mu$$

The second equality is observed by substituting ρ with μ . Finally,

$$\frac{1}{\sqrt{|g|}} \partial_\nu \sqrt{|g|} = \Gamma_{\mu\nu}^\mu \quad \square$$